

A multiplex panel for DNA stress and DNA repair capacity

Design document. Built outward from 8-OHdG to a panel that quantifies oxidative and alkylation damage, the cellular response to it, and the capacity of the pathways that protect against mutation.

0. Executive summary

8-OHdG (properly 8-oxo-7,8-dihydro-2'-deoxyguanosine, **8-oxodG**) is the most used oxidative DNA damage marker in the literature. It is also the single most artefact-prone analyte in the field, and on its own it answers the wrong question. A raised 8-oxodG cannot distinguish the two states a clinician actually cares about:

- **damage production is high, and repair is keeping up** — a flux state, largely benign
- **damage production is high, and repair is failing** — the mutagenic state

Both give the same 8-oxodG number. Separating them requires measuring repair in the same subject at the same time. That is the design principle of this panel.

The panel has three modules, because a single platform cannot carry both halves of the question. 8-oxodG is a small molecule quantified by mass spectrometry; MSH2 is a protein epitope quantified by antibody. No cartridge multiplexes those together. Each module is, however, genuinely multiplexed internally:

Module	What it measures	Platform	Matrix	Plex
A	oxidative, alkylation, halogenation and deamination adducts	LC-MS/MS, one MRM method	urine, DNA hydrolysate	18 analytes
A2	stress markers whose chemistry bars them from the MRM method	IHC, qPCR, flow, sequencing	tissue, blood, plasma	11 markers
B	cellular damage response	flow cytometry, one stained tube	PBMC	10 parameters
C	repair capacity	IHC + MSI/HRD + expression array	tissue, PBMC	9 tests + 83 targets

Headline readout is the Stress-to-Repair Index, defined in Section 5 as a difference of robust z-scores rather than a ratio, and reported *pathway-resolved* so that an abnormal result names the enzyme to follow up rather than just flagging a number.

Verification is built into the build. `src/dna_stress_panel.py` recomputes every mass-spectrometric transition from the molecular formula, checks every internal standard for a resolvable mass shift and an intact elemental backbone, tests the flow panel for spectral crowding, and confirms that every damage analyte has a named enzymatic partner in the repair module. It currently reports **0 failures across 11 computable transitions and 14 of 14 damage analytes coupled**. Three findings in Section 8 were produced by those checks and changed the design.

1. Why 8-OHdG needs company: the marker's three real problems

1.1 Measurement method changes the answer

This is not a minor calibration issue. A multi-laboratory comparison organised as a consensus exercise on urinary 8-oxodG found that **immunoassay (ELISA) returns systematically higher values than chromatographic methods, and the agreement between the two is poor** — a difference large enough that studies using different platforms are not comparable. The consensus recommendation was chromatographic quantification with isotope-labelled internal standards.

Design consequence. This panel specifies **LC-MS/MS with a stable-isotope-labelled internal standard for every analyte**. ELISA is excluded, not de-emphasised. Antibody cross-reactivity against structurally related urinary components is the accepted explanation for the positive bias, and no amount of local calibration fixes it.

1.2 The extraction step manufactures the analyte

Guanine oxidises during DNA isolation. Phenol-based extraction, iron contamination and air exposure all generate 8-oxodG that was not present in the cell. Reported tissue 8-oxodG levels in the older literature span more than an order of magnitude, and the spread is attributed largely to extraction chemistry rather than biology.

Design consequence. Section 6 specifies chelator-supplemented, phenol-free extraction, and the panel carries an **independent cross-check**: the FPG/OGG1-modified comet assay measures the same lesion class *inside intact cells*, where no extraction has occurred. Divergence between the LC-MS/MS result and the enzyme-modified comet result is the strongest available signal that the adduct number is an extraction artefact. No single-analyte assay can detect its own artefact this way.

1.3 Urinary and genomic 8-oxodG mean different things

Urinary 8-oxodG is excreted repair product. It is a **rate** — whole-body lesion turnover. Genomic 8-oxodG in DNA hydrolysate is a **steady-state level** — what is present right now. High repair activity *raises* the urinary number while *lowering* the genomic one.

Design consequence. The panel measures 8-oxodG in both matrices and never averages them. Their relationship is itself informative: high urinary with low genomic is competent repair under load; low urinary with high genomic is repair failure.

2. Module A — the adduct panel

Eighteen analytes in one liquid-chromatography tandem-mass-spectrometry method. All transitions below were **recomputed from molecular formula** by the build script, not transcribed. For 2'-deoxynucleosides the product ion is the protonated base, formed by loss of the intact sugar (C₅H₈O₃, 116.0473 Da); for the ribonucleoside the loss is C₅H₈O₄ (132.0423 Da).

2.1 Chemistry, and why each analyte is in the panel

Analyte	Lesion chemistry	Precursor → product	Why it earns a channel
8-oxodG	guanine C8 oxidation, DNA	284.10 → 168.05	The anchor. Most characterised, most comparable to prior literature
8-oxoGuo	guanine C8 oxidation, RNA	300.09 → 168.05	Independent axis. Urinary RNA-oxidation markers were associated with long-term mortality in newly diagnosed type 2 diabetes. Rises with mitochondrial and metabolic stress even when DNA oxidation is normal
8-oxoGua	free base	168.05 → 140.04	Excision-flux reporter. Released both by glycosylase action and by nucleotide-pool sanitisation. Rising alongside 8-oxodG means excision is <i>working</i>
FapyGua	ring-opened purine	170.07 → 152.06	Forms preferentially at low oxygen tension , where 8-oxoG formation is disfavoured. Omitting it systematically under-reports purine oxidation in hypoxic tissue
Tg	pyrimidine ring saturation	277.10 → 161.06	Substrate of NTHL1/NEIL1, not OGG1. Independent read on the non-OGG1 arm of base excision repair
5-hmdC	cytosine oxidation	258.11 → 142.06	Pyrimidine axis — but see the trap in 2.2
5-medC	methylation (not damage)	242.11 → 126.07	Not a lesion. Present solely to make 5-hmdC interpretable
O6-medG	O6 alkylation	282.12 → 166.07	The specific substrate of MGMT. The tightest analyte-to-enzyme pairing in the panel

Analyte	Lesion chemistry	Precursor → product	Why it earns a channel
N7-meGua	N7 alkylation, free base	166.07 → 149.05	Most abundant alkylation adduct; depurinates spontaneously so it appears in urine. Alkylating-exposure dosimeter
M1dG	malondialdehyde exocyclic adduct	304.10 → 188.06	Converts a lipid-peroxidation signal into a covalent DNA lesion
εdA	etheno-adenine	276.11 → 160.06	The only adenine-directed lesion; reports aldehyde and vinyl-monomer exposure
dG	unmodified	268.10 → 152.06	Denominator. Lesions reported per 10 ⁶ dG. Also confirms hydrolysis went to completion
Creatinine	normaliser	114.07 → 44.05	Mandatory for spot urine
8-iso-PGF2 α	lipid peroxidation	353.23 → 193.12 (negative)	Separates systemic oxidative stress from DNA-specific damage. Pooled reference IQR 0.18–0.40 µg/g creatinine by chemical methods in adults with BMI under 25
cdA	8,5'-cyclo-dA, tandem cyclopurine	250.09 → 164.06	Oxidative lesion repaired by NER , not BER . See 2.5
cdG	8,5'-cyclo-dG	266.09 → 180.05	Guanine counterpart; R and S diastereomers must be resolved
5-Cl-dC	chlorinated cytosine	262.06 → 146.01	Myeloperoxidase/HOCl product. Neutrophil-driven inflammation, chemically distinct from oxidation and nitration
dU	uracil in DNA	229.08 → 113.03	Cytosine deamination and dUTP misincorporation. A third damage mechanism: neither oxidative nor alkylative

2.2 The 5-hmdC trap

5-hydroxymethyl-dC is **both** an oxidation product of cytosine **and** a programmed epigenetic mark written deliberately by TET enzymes. Scoring it as damage without reference to its precursor pool will misread normal epigenetic remodelling as oxidative injury. This is why 5-medC occupies a channel: 5-hmdC is reported as a fraction of 5-medC, and the panel carries TDG and SMUG1 in the expression module so that programmed turnover can be distinguished from oxidative chemistry.

2.3 Why the panel is chemically non-redundant

The analytes are deliberately spread across **different base targets and different oxidation chemistries**, because reactive species have different preferences:

- purine oxidation, closed-ring: 8-oxodG, 8-oxoGuo, 8-oxoGua
- purine oxidation, ring-opened, oxygen-tension-dependent: FapyGua
- pyrimidine oxidation: 5-hmdC, Tg
- alkylation: O6-medG, N7-meGua
- aldehyde-derived exocyclic adducts: M1dG, εdA
- adenine-directed: εdA

A panel of ten guanine-oxidation markers would be ten measurements of one thing. This one spans four lesion chemistries on three bases across DNA and RNA.

2.4 Two flagged ion overlaps

The build script identified two pairs sharing a product ion:

- **8-oxodG and 8-oxoGuo both fragment to 168.05.** This is expected and correct: both release the same protonated 8-oxoguanine base. Precursors differ by 16 Da, so the pair is fully resolved in the first quadrupole.
- **FapyGua and dG both report a product near 152.06.** Precursors are 170.07 and 268.10 — resolved, but retention-time separation should be confirmed since FapyGua is the ring-opened analogue of the same nucleobase.

Neither is a collision. No analyte pair shares *both* precursor and product, which the script verifies explicitly.

2.5 The cyclopurines, and why they are the most useful addition

8,5'-cyclo-2'-deoxyadenosine and its guanine counterpart form when a hydroxyl radical attacks the sugar and the resulting radical cyclises between C5' and C8. That covalent bond between sugar and base is the whole point:

- It **distorts the duplex**, so the lesion is recognised as bulky helix-damaging injury.
- It means these lesions are **repaired by nucleotide excision repair rather than base excision repair**, unlike every other oxidative lesion in the panel.
- It also means **the glycosidic bond cannot be cleaved**. The build script's neutral-loss check correctly refuses to compute a product ion for these two analytes for exactly this reason, and says so in its output.

The design value is considerable. 8-oxodG and cdA are produced by the **same oxidant** but cleared by **different pathways**. Measuring both lets a single oxidative stress interrogate two repair systems at once. If 8-oxodG is normal while cdA is elevated, the oxidant load is ordinary and the NER arm is the problem — a conclusion no guanine-oxidation marker can reach on its own. These lesions block polymerases, inhibit gene expression, and accumulate in tissue in association with disease.

2.6 Four damage chemistries, not one

With these additions the adduct module spans mechanistically distinct injury types, each with its own biological source:

Chemistry	Analytes	Source
Oxidation, closed-ring purine	8-oxodG, 8-oxoGuo, 8-oxoGua	general reactive oxygen species
Oxidation, ring-opened purine	FapyGua	ROS at low oxygen tension
Oxidation, tandem cyclisation	cdA, cdG	hydroxyl radical specifically
Oxidation, pyrimidine	5-hmdC, Tg	ROS
Nitration	8-nitroguanine (Module A2)	peroxynitrite, chronic inflammation
Halogenation	5-Cl-dC	myeloperoxidase/HOCl, neutrophils
Alkylation	O6-medG, N7-meGua	nitrosamines, tobacco, chemotherapy
Aldehyde exocyclic	M1dG, εdA	lipid peroxidation
Deamination	dU	spontaneous, folate deficiency, antifolates

This matters because the chemistries point at different causes. Halogenation implicates neutrophils. Nitration implicates peroxynitrite and chronic inflammation. Deamination implicates folate status or thymidylate-synthase inhibition. Alkylation implicates exposure. A panel confined to guanine oxidation cannot distinguish any of these.

2A. Module A2 — markers that cannot go in the mass-spectrometry method

Eleven markers in `data/dna_stress_other_markers.csv`. Each is excluded from Module A for a **chemical or compartmental reason**, not an arbitrary one, and several are as clinically familiar as 8-OHdG itself.

2A.1 8-nitroguanine — the nitrative axis

Nitration is a damage chemistry the adduct panel otherwise misses entirely. 8-nitroguanine is a mutagenic lesion formed during chronic inflammation via reactive nitrogen species. It has been demonstrated at sites of carcinogenesis, accumulates in cancer-prone inflammatory disease caused by several pathogens including human papillomavirus and Epstein-Barr virus, and strong formation in tumour tissue has been closely associated with poor prognosis. Given that chronic inflammation is estimated to account for roughly a quarter of cancer cases, an inflammation-specific DNA lesion earns its place.

Why it is not in the MRM method: 8-nitroguanine is chemically labile and depurinates rapidly, so immunohistochemistry with a specific antibody is the established detection route. The marker's own instability dictates

its platform.

2A.2 Abasic sites — measuring the bottleneck directly

An abasic site is the obligate intermediate between glycosylase excision and gap filling, and it is **more mutagenic than the base lesion it replaced**. Section 4.2 argues that high glycosylase activity with low APE1 is worse than low glycosylase activity; abasic site burden is how you measure that state instead of inferring it. Quantified by aldehyde-reactive probe, with methoxyamine binding abasic sites covalently.

2A.3 The mitochondrial genome — a separate compartment

Three markers cover it: copy number, the 4977 bp common deletion, and polymerase-blocking lesion frequency by long-amplicon qPCR. Mitochondrial DNA **lacks nucleotide excision repair** and sits adjacent to the electron transport chain, so it accumulates oxidative damage faster than nuclear DNA. The common deletion has been reported at higher levels in the blood of breast cancer patients with concurrently lower mitochondrial DNA content and higher oxidative damage markers.

A nuclear DNA hydrolysate does not report this compartment at all. Copy number is read only alongside deletion burden, because copy number alone moves with both mitochondrial biogenesis and mitochondrial loss and is ambiguous by itself.

2A.4 Telomere length — a different time constant

Telomeric DNA is guanine-rich, disproportionately sensitive to oxidation, and refractory to repair. Psychological stress has been associated with higher oxidative stress, lower telomerase activity and shorter telomeres, with the highest-stress group shorter by the equivalent of at least a decade of additional ageing.

Its role here is **temporal**. Adducts integrate hours to days. Telomere length integrates months to years. Including both separates an acute insult from a chronic burden, which no single-timescale marker can do. The panel pairs it with TERT and TERF2 transcripts, since telomerase activity and telomere length answer different questions.

2A.5 Mutation outcome — what the whole panel is actually about

Every other marker is a lesion or an enzyme. These two are fixed mutations:

- **PIG-A mutant frequency**, by flow cytometry for loss of GPI-anchored proteins on erythrocytes. Reported background is roughly 2.9 to 5.56 per million mutant erythrocytes in healthy subjects, with increases after cigarette smoking, radiotherapy and occupational exposures including lead. A low, stable background is what makes induced mutation detectable.
- **Error-corrected (duplex) sequencing**, which returns mutation frequency *and* spectrum — an outcome measure and a mechanism attribution in one assay. Inter-individual variability in clonally expanded mutations is a known analytical challenge.

A panel that measures damage and repair but never mutation is measuring inputs and machinery while declining to check the output.

3. Module B — the cellular damage response

Ten parameters on one stained tube, three lasers. The channel plan is in `data/ddr_flow_panel.csv`; the spectral check output is in `results/moduleB_spectral_check.csv`.

3.1 Two channels are not optional

Cleaved caspase-3 (BV605). Apoptotic cells generate pan-nuclear γ H2AX from programmed DNA fragmentation. That signal is not repair-relevant damage, and a panel without an apoptosis gate reports cell death as genotoxic stress. Work correlating γ H2AX with radiation-induced apoptosis in lymphocytes found a **negative** correlation between residual γ H2AX and apoptosis at 48 hours, interpreted as individuals less efficient at clearing damaged cells retaining higher

basal and residual damage. Whether a cell is dying or repairing changes the meaning of its γ H2AX entirely.

DNA content (DAPI). γ H2AX and RAD51 are strongly S/G2-dependent. Uncorrected foci counts track proliferation rather than damage. Every foci measurement in this panel is reported cell-cycle-resolved.

3.2 The damage and repair channels

Target	Reads	Fluorochrome	Why
γ H2AX (Ser139)	double-strand breaks	AF488	Best-characterised cellular damage marker; used as a pharmacodynamic endpoint in trials
53BP1 foci	DSB recognition	PE	Co-localisation with γ H2AX upgrades a signal to a genuine break
pATM (Ser1981)	apical DSB kinase	PE-Dazzle594	Distinguishes damage being <i>signalled</i> from damage being ignored
pCHK2 (Thr68)	checkpoint arm	PerCP-Cy5.5	High γ H2AX with low pCHK2 is checkpoint escape — a pro-mutagenic state
pKAP1 (Ser824)	DSB-specific signalling	APC-Fire750	More DSB-specific than γ H2AX; not induced by replication stress alone
RAD51 foci	HR execution	APC	Functional HR, as distinct from the genomic HRD scar
PAR polymer	PARP1 activity	AF700	Catalytic output, the functional counterpart of PARP1 transcript
Ki-67	proliferation covariate	BV711	A damaged quiescent cell and a damaged cycling cell mean different things

γ H2AX is **sensitive but not specific** — it rises in replication stress and apoptosis as well as at true breaks. The panel therefore never reads it alone: 53BP1 co-localisation, pKAP1 specificity, and the caspase-3 gate are what make a γ H2AX number interpretable.

3.3 Spectral finding

The script flagged one spillover-critical adjacency: **53BP1 (PE, 578 nm) against pATM (PE-Dazzle594, 610 nm)**, a 32 nm gap on the same 488 nm laser. This pair requires single-stain compensation controls and, if the instrument allows, reassignment of pATM to a red-laser channel. Reported as a build finding rather than silently accepted.

3.4 Cytogenetic endpoints — the only prospectively validated damage measure

Run from the same blood draw, separate culture. These are in `data/ddr_cytogenetic_assays.csv`.

The **cytokinesis-block micronucleus assay** carries something no other endpoint in this panel has: **prospective validation against cancer incidence in humans**. In a 6718-subject cohort across ten countries and twenty laboratories, subjects in the medium and high micronucleus-frequency groups showed increased all-cancer incidence (relative risk 1.84, 95% CI 1.28–2.66 for medium; 1.53, 95% CI 1.04–2.25 for high) and reduced cancer-free survival. The association held across national cohorts and major cancer sites.

Two design points follow. First, the micronucleus assay is the **anchor for clinical meaning** in the whole panel; the adduct measurements are mechanism, this is outcome. Second, that study standardised subjects by **percentile of the distribution within each laboratory**, precisely because inter-laboratory variability is large. Any deployment of this panel must band micronucleus results against its own laboratory distribution, not against a published absolute threshold.

The **alkaline comet assay** likewise has pooled international baseline frequencies and characterised confounders available from large database-comparison efforts; use those for banding rather than inventing local cut-offs.

The **ex-vivo challenge comet assay** is what converts the panel from a snapshot to a capacity measurement: give PBMCs a calibrated dose, measure residual damage at 2 and 24 hours, and read repair kinetics directly. Residual damage at 24 hours is the functional deficiency readout.

4. Module C — repair capacity, with mismatch repair as the anchor

4.1 The clinically established tests

Nine tests in `data/repair_clinical_tests.csv`. These are the parts of the panel that already have regulatory and guideline standing, and mismatch repair is where that standing is strongest.

MMR protein immunohistochemistry — MLH1, MSH2, MSH6, PMS2. Loss of nuclear staining with retained internal control indicates deficient MMR. This four-antibody panel is routine diagnostic practice, embedded both in Lynch-syndrome screening and in checkpoint-inhibitor eligibility.

The heterodimer rule, which the panel enforces. These four proteins work as two obligate pairs. MSH2 partners MSH6; MLH1 partners PMS2. **MSH6 loss can be secondary to MSH2 loss, and PMS2 loss secondary to MLH1 loss.** Scoring the four markers independently over-calls primary mutation events. The panel specifies pair-aware interpretation. Expression data reflecting MMR-protein status has been examined in checkpoint-inhibitor-treated cohorts outside colorectal cancer, which is a reminder that the four-marker readout travels beyond its original tumour setting and needs its interpretation rules to travel with it.

MSI by PCR (BAT-25, BAT-26, NR-21, NR-24, MONO-27) is the orthogonal confirmation. It carries a specific documented trap: **minimal microsatellite shift occurs in MSI-high tumours — notably endometrial — and is a recognised cause of false-negative interpretation.** The panel specifies matched normal tissue and an explicit analyst alert for small shifts, rather than relying on automated peak calling.

MGMT promoter methylation is the alkylation-repair readout and is predictive for alkylating chemotherapy benefit, most established in glioblastoma where temozolomide has been compared against radiotherapy regimens in randomised trials. The panel requires the **quantitative** methylation value, not only the binary call, because the methylated/unmethylated cut-off is assay-dependent and the intermediate zone is genuinely ambiguous.

HRD score (loss of heterozygosity + telomeric allelic imbalance + large-scale state transitions) reports homologous-recombination status for PARP-inhibitor decisions. The important limitation, and the reason Module B carries RAD51 foci: **a genomic scar is a historical record, not a statement about present-day function.** A tumour can carry a high HRD score and have restored RAD51 loading — that is a PARP-inhibitor resistance phenotype, and the scar score alone will misclassify it. Available HRD assays differ and each requires validation on clinical samples before clinical use.

POLE / POLD1 exonuclease domain sequencing covers the mutation-protection pathway that mismatch repair testing cannot see, and it is the single most important addition to the repair module.

Why an MMR-only panel is not enough. Polymerases ϵ and δ replicate the genome and proofread their own work through exonuclease domains. Exonuclease domain mutations produce an **ultramutated** tumour — sometimes exceeding a million base substitutions — that is nonetheless **apparently microsatellite stable**. An MMR immunohistochemistry and MSI workup reports such a tumour as normal. The most extreme mutator phenotype in human cancer is invisible to the tests that define the mismatch repair readout. Germline exonuclease domain mutations in these genes define polymerase proofreading-associated polyposis, with high penetrance and dominant inheritance. Proofreading and mismatch repair defects can also co-occur, and that combination produces the highest mutation rates observed — somatic POLE exonuclease mutations arising on a background of biallelic mismatch repair deficiency have been reported with tumour mutation rates of 237 and 123 per megabase.

Framing the problem as mismatch repair *or other mutation-protection pathways* is the right framing, and proofreading is the pathway it points at. The panel therefore treats replication fidelity as a **two-layer system**: the polymerase proofreads as it goes, and mismatch repair corrects what proofreading missed. Testing only the second layer leaves the first unexamined, and the first is where the largest mutation burdens come from. POLE, POLD1 and PCNA are in the expression panel under pathway code PR00F.

Mutational signature deconvolution is the only element that attributes observed mutations to a *specific failed pathway* — MMR deficiency, HR deficiency, oxidative damage and alkylation each leave distinguishable single-base-substitution signatures. Tumour mutational burden, by contrast, is a consequence of accumulated repair failure and is mechanistically uninformative on its own.

4.2 The multiplexed expression panel

83 targets across eleven repair pathways plus six normaliser candidates, in `data/repair_expression_panel.csv`. Two tiers, sized by the build script against real plate formats:

Tier	Targets	Format	Spare wells
Core clinical	46	48-plex qPCR array	2
Extended research	83	96-plex array	13

Pathway coverage:

Code	Pathway	Genes
BER	base excision repair	20
HR	homologous recombination	11
SIG	damage signalling	9
NER	nucleotide excision repair	8
MMR	mismatch repair	7
NHEJ	non-homologous end joining	7
ICL	interstrand crosslink repair	4
DR	direct damage reversal	3
TLS	translesion synthesis	3
PROOF	polymerase proofreading	3
TEL	telomere maintenance	2

Base excision repair carries the most genes deliberately: it is the pathway that handles the panel's anchor lesion, and it has enough sequential steps that knowing *which* step is limiting matters. A glycosylase excises the base; APE1 incises the resulting abasic site; POLB fills the gap; XRCC1 scaffolds; LIG3 seals. **High glycosylase activity with low APE1 is worse than low glycosylase activity**, because it converts a modified base into an abasic site, which is more mutagenic than the lesion it replaced. A pathway-level score would miss that; a step-resolved panel sees it.

Translesion synthesis is included for the same reason with opposite sign. REV3L and REV1 provide damage **tolerance**, not repair — they let replication proceed past a lesion at the cost of fidelity. **High TLS with low repair is the actively mutagenic configuration**, and it looks like coping if you only measure whether replication completed.

Normalisation uses the geometric mean of stability-selected housekeepers from the six-gene candidate set, not a single reference gene. B2M is included as a candidate but flagged: it is interferon-responsive, so inflammation moves it, and inflammation is correlated with oxidative stress — exactly the confounding structure that would bias this panel.

4.3 The damage-to-repair coupling, which is the point

Every damage analyte is mapped to the enzymes that handle it. The build script verifies this mapping and reports **14 of 14 damage analytes coupled**:

Damage analyte	Pathway	Enzymatic partners in panel
8-oxodG	BER	OGG1, MUTYH
8-oxoGuo	BER	NUDT1 (MTH1)
8-oxoGua	BER	NUDT1 (MTH1)
FapyGua	BER	NEIL1, NEIL2
5-hmdC	BER	SMUG1, TDG
Tg	BER	NTHL1
O6-medG	DR	MGMT

Damage analyte	Pathway	Enzymatic partners in panel
N7-meGua	BER	MPG
M1dG	NER	XPC, ERCC1, ERCC4
εdA	NER	XPC, ERCC1, ERCC4
cdA	NER	XPA, XPC, ERCC1, ERCC2, ERCC4, ERCC5, DDB2
cdG	NER	XPA, XPC, ERCC1, ERCC2, ERCC4, ERCC5, DDB2
5-Cl-dC	BER	SMUG1, TDG
dU	BER	UNG, SMUG1, DUT, TYMS

The RNA analyte is the interesting case. **Oxidised RNA is not repaired in place** — there is no RNA equivalent of base excision repair. Its only enzymatic defence is MTH1 hydrolysing 8-oxo-GTP out of the nucleotide pool before incorporation. So 8-oxoGuo couples to NUDT1 and to nothing else, and that is a statement about biology rather than a gap in the panel.

5. Quantification

5.1 Per-analyte normalisation

- **Spot urine:** divide by creatinine. Non-negotiable — unnormalised spot-urine concentrations are not comparable between subjects. Report as nmol/mol creatinine.
- **24-hour urine:** report as absolute excretion per 24 h, and as per kg body weight.
- **DNA hydrolysate:** report as lesions per 10⁶ dG, using the measured dG channel as denominator.
- **Flow:** report as median fluorescence intensity ratio to unstimulated control, and as percentage of cells above a control-defined threshold, both cell-cycle-gated.

5.2 Robust z-scoring on the log scale

Adduct distributions are right-skewed and outlier-prone. The panel scores on the log scale against a laboratory-specific reference **median and interquartile range**, not mean and standard deviation:

$$z_i = (\ln(x_i) - \ln(\text{median}_{\text{ref},i})) / (\text{IQR}_{\text{ref},i} / 1.349)$$

The 1.349 divisor converts an interquartile range into a standard-deviation equivalent for a normal distribution, so the score remains interpretable on the familiar z scale while resisting the outliers that wreck a parametric z.

5.3 Precision weighting

Analytes are not measured equally well. An analyte at 25% coefficient of variation should not carry the same weight in a composite as one at 5%. Weights are inverse to log-scale measurement variance:

$$w_i = (1 / \ln(1+CV_i)^2) / \sum_j (1 / \ln(1+CV_j)^2)$$

In the worked example this gives 8-oxodG (6% CV) a weight of 0.392 and M1dG (15% CV) a weight of 0.068 — a factor of six, reflecting how much each measurement can be trusted.

5.4 The three composite indices

- **ODI, Oxidative Damage Index** — precision-weighted mean of Module A lesion z-scores.
- **RCI, Repair Capacity Index** — reported *per pathway* (RCI_MMR, RCI_BER, RCI_NER, RCI_HR, RCI_NHEJ, RCI_DR) and as a mean across pathways.
- **SRI, Stress-to-Repair Index** — the headline number.

5.5 Why the headline index is a difference and not a ratio

The intuitive construction is damage divided by repair. It is wrong, and the reason is worth stating because it is a trap that has appeared in published composite scores.

A z-scored denominator **crosses zero by construction** — that is what z-scoring means. So ODI/RCI is undefined at the population median and explodes in a neighbourhood around it, giving enormous scores to subjects who are close to average. The panel instead uses

$$\text{SRI} = z(\text{ODI}) - z(\text{RCI})$$

On the log scale a difference is exactly what a ratio is reaching for ($\ln(a/b) = \ln a - \ln b$), and it is finite everywhere. Interpretation:

SRI	State
> +0.5	decompensated — damage exceeds repair capacity; the mutagenic risk state
−0.5 to +0.5	compensated — damage matched by repair
< −0.5	repair reserve

5.6 What makes the result actionable

Run `python3 src/dna_stress_panel.py` for the worked arithmetic on illustrative values. In that example $\text{SRI} = +1.51$, decompensated. The composite is driven by $\text{RCI_BER} (-1.45)$ and $\text{RCI_DR} (-0.90)$, and the coupling table names OGG1, MUTHYH, NUDT1 and MGMT as the specific enzymes to follow up. **A single scalar damage score would have flagged the subject without localising the defect.** Pathway resolution is the difference between a result and a next step.

The numbers in that example are invented to demonstrate the arithmetic. They are not measurements and carry no reference-interval authority.

6. Pre-analytics, where this panel is most likely to fail

Pre-analytical error dominates every other error source for these analytes. This section is not boilerplate.

6.1 Urine

- First morning void, or timed 24-hour collection. Do not mix designs within a study.
- Aliquot and freeze at -80°C within 2 hours.
- **Avoid repeated freeze-thaw.** Aliquot at first freeze into single-use volumes.
- Measure creatinine on the same aliquot as the adducts.
- Record BMI, smoking status and time of collection. All three move these analytes; urinary 8-isoprostane in particular shows a significant positive association with BMI.

6.2 Blood and DNA extraction — the critical path

- EDTA tubes; isolate PBMCs within 2 hours.
- **Extraction chemistry determines the 8-oxodG result.** Use a chelator-supplemented, phenol-free method (salting-out or anion-exchange). Add a metal chelator such as deferoxamine to suppress Fenton chemistry during lysis. Minimise air exposure and work cold.
- Enzymatic hydrolysis to nucleosides, not acid hydrolysis: acid conditions oxidise guanine and destroy the ring-opened lesions.
- Run the FPG/OGG1-modified comet assay on cells from the same draw as the artefact cross-check described in 1.2.

6.3 Tissue

FFPE is acceptable for MMR immunohistochemistry, MSI, MGMT methylation and HRD.

FFPE is not acceptable for oxidative adduct quantification. Formalin fixation and prolonged block storage oxidise DNA. Adduct measurements require fresh or snap-frozen tissue. This restriction is why Module A and Module C can run on different matrices from the same subject and must not be forced onto one specimen type.

7. Validation plan before any clinical use

Element	Target
Linearity	≥ 5 calibration levels, $r^2 \geq 0.99$, across the expected physiological range
Lower limit of quantification	$CV \leq 20\%$ and accuracy 80–120% at the LLOQ
Intra-assay precision	$CV \leq 10\%$ for Module A nucleosides; $\leq 15\%$ for free bases
Inter-assay precision	$CV \leq 15\%$ across ≥ 20 independent runs
Matrix effect	post-column infusion assessment for every analyte in every matrix
Internal standard M+3 bleed	quantify the analyte's natural M+3 contribution into the IS channel for the six analytes flagged at +3 nominal shift
Carryover	$\leq 20\%$ of LLOQ in a blank following the highest calibrator
Reference interval	≥ 120 healthy subjects per stratum, sex- and age-stratified, per CLSI-style design
Stability	freeze-thaw $\times 3$, bench-top 24 h, long-term -80°C at 6 and 12 months
Orthogonal confirmation	Module A 8-oxodG against FPG-modified comet on paired samples
Laboratory banding	establish local percentile distributions for micronucleus and comet endpoints rather than adopting published absolute cut-offs

8. What the build checks found

Two findings from the automated verification changed the design rather than being written up after the fact.

The internal-standard threshold was initially wrong. The first implementation required an exact mass shift of ≥ 3.0 Da and failed four internal standards whose shifts were 2.991–2.997 Da. Those are [13C1, 15N2] and [15N3] labels — **+3 nominal** mass shifts, fully resolvable on a unit-resolution quadrupole. The check was testing exact mass where the instrument resolves nominal mass. It now computes the integer nominal shift, requires ≥ 3 , and issues an advisory (PASS_VERIFY_M3) at exactly +3 so that M+3 isotopic bleed-through from the analyte into the standard's channel is verified during method development. Six analytes carry that advisory; it appears in the validation plan in Section 7.

Two internal standards were upgraded. FapyGua and M1dG are both guanine-derived, so [15N5]-labelled precursors are commercially available. Both were moved from +3 to +5 nominal shift, removing the bleed-through concern entirely for those two channels.

One spectral adjacency was flagged and left visible. 53BP1 (PE) and pATM (PE-Dazzle594) sit 32 nm apart on the same laser. Rather than quietly reassigning it, the build reports it as spillover-critical so that compensation controls are treated as mandatory for that pair.

A validation rule refused to apply itself, correctly. The glycosidic neutral-loss check computes a product ion by subtracting the intact sugar from the precursor. For the two cyclopurines it declines to do so and reports the reason: the C5'-C8 covalent bond that defines the lesion is precisely what prevents the sugar leaving as a neutral fragment. The validator encoding that distinction is what stops a plausible-looking but chemically impossible transition from being accepted for cdA and cdG.

One coupling gap turned out to be biology. The first coupling run reported 8-oxoGuo as uncoupled. That was correct as written but incomplete: MTH1 sanitises 8-oxo-GTP as well as 8-oxo-dGTP, so the RNA analyte does have an enzymatic partner. The annotation was corrected, and the underlying point — oxidised RNA is not repaired in place — is now stated explicitly in Section 4.3 instead of appearing as a missing entry.

9. Honest limitations

- **No composite index in this panel is clinically validated.** ODI, RCI and SRI are defined here as a coherent scoring framework, not as validated clinical instruments. They require prospective outcome validation before any clinical decision rests on them. The individual components differ enormously in maturity: MMR immunohistochemistry and MSI are routine diagnostics, the micronucleus assay has prospective cancer-incidence data, and most Module A adducts beyond 8-oxodG are research-grade.
- **Reference intervals are laboratory-specific.** Published absolute values for these analytes are not transferable between laboratories, which is the central lesson of the standardisation literature cited throughout.
- **Transcript abundance is not enzyme activity.** Module C's expression panel measures message, not function. That is why the design pairs transcripts with functional readouts wherever one exists: PARP1 with PAR polymer, RAD51 transcript with RAD51 foci, CHEK2 with pCHK2, H2AX with γH2AX, MGMT with promoter methylation. Where no functional counterpart exists, the transcript result is weaker evidence and should be reported as such.
- **The ex-vivo challenge assay measures lymphocyte repair capacity,** which is a proxy for the tissue of interest, not a measurement of it.
- **Mutational signature attribution needs mutation count.** Exome-scale data give unstable signature fits; the attribution layer in Section 4.1 needs genome-scale sequencing to be reliable.
- **Module A2 spans wildly different maturity levels.** Telomere length and mitochondrial copy number have large literatures and poor standardisation; PIG-A has a well-characterised background frequency; error-corrected sequencing is still being established for human biomonitoring. The nitrative and halogenative markers are research-grade.
- **Cell-free DNA is context, not a damage measure.** It reports that cells died, not that their genomes were damaged, and leukocyte lysis during sample handling contaminates the signal. It is included with that caveat attached rather than scored as damage.
- **Cost and throughput are not addressed here.** A three-module panel requiring LC-MS/MS, flow cytometry, immunohistochemistry, fragment analysis and an expression array is a research instrument. Deciding which modules survive into a deployable assay is a separate exercise from designing the complete one.

10. Files

data/dna_damage_adducts.csv	18 Module A analytes: formula, transition, internal standard
data/dna_stress_other_markers.csv	11 Module A2 markers on other platforms
data/ddr_flow_panel.csv	10 Module B flow parameters with laser/emission assignment
data/ddr_cytogenetic_assays.csv	5 cytogenetic and comet endpoints
data/repair_clinical_tests.csv	9 clinically established repair tests with pitfalls
data/repair_expression_panel.csv	83 repair genes across 11 pathways, 2 tiers
src/dna_stress_panel.py	builder, validator and scoring framework
src/build_report_pdf.py	renders the report to PDF
results/	generated verification output

Sources

Literature located through PubMed. Key references, with DOI links:

- Urinary 8-oxodG measurement consensus and the ELISA-versus-chromatography discrepancy — Cooke *et al.*, *FASEB J* ([DOI](#))
- Micronucleus frequency as a prospectively validated cancer-risk biomarker, 6718 subjects — Bonassi *et al.*, *Carcinogenesis* 2006 ([DOI](#))
- Genetic damage biomarkers for cancer epidemiology — Fenech, *Toxicology* 2002 ([DOI](#))
- Comet assay in human biomonitoring, international database comparison and baseline frequencies — *Mutat Res Rev Mutat Res* ([DOI](#)) and ([DOI](#))
- Urinary nucleic-acid oxidation markers and long-term mortality in type 2 diabetes — *Diabetes Care* ([DOI](#)); RNA oxidation follow-up ([DOI](#))
- γ H2AX as a quantitative DSB biomarker — *Aging* ([DOI](#)); γ H2AX and other histone modifications in the clinic — *Biochim Biophys Acta* ([DOI](#))
- γ H2AX, apoptosis and radiosensitivity in patient lymphocytes — *Int J Radiat Biol* ([DOI](#)); DSB repair capacity and normal-tissue toxicity — *J Radiat Res* ([DOI](#))
- OGG1-driven base excision repair in disease — *Curr Opin Allergy Clin Immunol* ([DOI](#))
- MMR protein expression and checkpoint-inhibitor treated disease — *J Cancer Res Clin Oncol* ([DOI](#))
- Minimal microsatellite shift as an MSI interpretation pitfall — *Mod Pathol* ([DOI](#))
- Homologous recombination deficiency biomarkers and assay limitations — *J Pers Med* ([DOI](#))
- Temozolomide versus radiotherapy in elderly glioblastoma, the MGMT-predictive setting — *Lancet Oncol* ([DOI](#)); glioblastoma overview ([DOI](#))
- Urinary 8-isoprostane reference values and BMI association, systematic review and meta-analysis — *Toxicol Lett* ([DOI](#))

Markers added in the expansion:

- 8-nitroguanine in inflammation-related carcinogenesis — Hiraku, *Environ Health Prev Med* ([DOI](#))
- 8,5'-cyclopurine-2'-deoxynucleosides: formation, measurement, NER-dependent repair and biological effects — Jaruga & Dizdaroglu, *DNA Repair* ([DOI](#))
- Polymerase ϵ and δ proofreading mutations defining a hypermutated, microsatellite-stable cancer class — Briggs & Tomlinson, *J Pathol* ([DOI](#)); germline PMS2 with somatic POLE exonuclease mutation and the resulting mutation rates — *J Pathol* ([DOI](#))
- Mitochondrial common deletion elevated in blood with oxidative stress — *Mitochondrion* ([DOI](#))
- Telomere shortening with life stress, oxidative stress and telomerase activity — Epel *et al.*, *PNAS* ([DOI](#))
- PIG-A gene mutation assay in human biomonitoring, including background mutant frequencies — *Environ Mol Mutagen* ([DOI](#))
- Duplex (error-corrected) sequencing for in-vivo mutation measurement — *Biol Reprod* ([DOI](#))
- Direct detection and quantification of abasic sites, including methoxyamine binding — *Nucl Med Biol* ([DOI](#))

Mass-spectrometric transitions in Section 2 are computed from molecular formula by `src/dna_stress_panel.py`. Nominal transitions for free bases and for 8-isoprostane are literature-typical values marked `lit.` in the build output and must be confirmed on the local instrument during method development.